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Cloud-Edge Collaborative Large Model Services: Challenges and Solutions

Yanghe Pan, Zhou Su, Yuntao Wang, Shaolong Guo, Han Liu, Ruidong Li, and Yuan Wu

***Abstract*—The rapid development of large models such as GPT-4 and Midjourney has spawned worldwide attention in various fields. To practically deploy large model services in downstream tasks, cloud-edge collaboration mechanism offers an appealing solution by seamlessly sharing fresh data, knowledge, and resources between the cloud and distributed edge nodes. Offloading the fine-tuning process of large models to edge networks through federated learning (FL) to implement efficient and privacy-preserving large model services has become an emerging research direction. However, there exist challenges in terms of privacy leakages, data heterogeneity, and edge resource constraints in delivering cloud-edge collaborative large model services. In this paper, we propose an innovative cloud-edge collaborative framework for federated large model training and deployment with reduced communication overheads and enhanced scalability. Within this framework, we further devise a data-model-driven privacy protection mechanism, a lightweight reciprocal knowledge transfer approach, and an optimal and reliable edge node recruitment strategy to fully harness edge data/knowledge/resources with enhanced privacy protection under high heterogeneity. Through an experimental case study, we validate the effectiveness of the proposed framework and solutions. Finally, we outline significant future research directions in this evolving field.**

***Index Terms*—large model, artificial intelligence, cloud-edge computing**

I. Introduction

Large models refer to artificial intelligence (AI) models with an extremely large number of parameters (i.e., typically larger than 1B), possessing superior understanding and generative capabilities compared to conventional AI models [1]. The large models, such as GPT-4 and Midjourney, not only demonstrate remarkable efficiency in fields such as natural language processing (NLP) and text-to-image generation, but also pave the way for innovative applications such as metaverse, cybertwins, and smart city [2]. As the most well-known large model, ChatGPT is anticipated to reach a revenue generation of $1 billion by 2024 with more than 180.5 million users monthly and 100 million weekly active participants. The emergence of large models leads a new era of technological innovation and broadens the horizons for AI applications.

Nevertheless, in conventional cloud-based large model service framework, to train a domain-specific large model for specific downstream tasks, it usually requires transmitting massive sensitive data samples from edge nodes to the central cloud. Thereby, it suffers from high latency and privacy concerns. The edge nodes serve as distributed computing resources located at the periphery of the network, such as edge computing platforms. Through cloud-edge collaboration, the computation/data/knowledge resources of cloud servers and edge nodes can be fully harnessed for effective large model service offering, while ensuring the privacy of local data within edge devices [3]. For instance, SECO deploys a large model in the edge computing platform of the industrial Internet of Things (IoT), enabling effective analysis of device data and providing guidance to operators in troubleshooting[1]. Specifically, the cloud-edge collaborative framework for large model services has the following advantages.

- *Collaborative privacy protection*. In a cloud-edge collaborative framework, the utilization of federated learning (FL) techniques allows large model fine-tuning while empowering collaborative privacy protection. Specifically, edge nodes can collaboratively train a domain-specific large model to support specific downstream tasks, thereby enabling sensitive user/business data to be processed locally on edge nodes.
- *Low-latency customized services*. By deploying edge servers closer to end-users, the cloud-edge collaboration framework can significantly reduce the communication latency between the users and the server. The proximity ensures faster responses to user inputs, which is essential for real-time and interactive large model services, thereby significantly enhancing the quality of experience (QoE) of users.
- *Dynamic decision optimization.* The foundation model deployed on the cloud provides comprehensively global-level understanding and decision-making for user inputs, while the specific large model deployed on the edge optimizes the global decision-making according to specific scenarios. Through the cloud-edge coordinated model combination, it can optimize decision-making process in an adaptive manner under the dynamic environment.

However, there are still efficiency and security issues in fully implementing large model services in the cloud-edge collaborative framework.

- *Limited computation and communication resources of edge nodes*. The limited resources of edge nodes pose a significant challenge in handling computation-intensive tasks. Moreover, bandwidth limitations further prolong

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[1]https://www.seco.com/blog/is-the-future-of-generative-ai-at-the-edge/

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communication delays. Therefore, it is essential to mitigate the communication and computational overheads by integrating with emerging techniques such as knowledge distillation.

- *Data heterogeneity among edge nodes*. Edge nodes possess diverse data in terms of quantity and quality. The non-independent and non-identically distributed (non-IID) nature of data among edge nodes can lead to biased large models. Meanwhile, it is impractical to recruit all edge nodes and assess their contributions during the large model fine-tuning process. Hence, it is crucial to pre-recruit the optimal edge nodes whose data aligns with the downstream task.
- *Privacy leakage of cloud-edge collaborative large model services*. The data memorization phenomenon in large models has been an emerging privacy concern [4], [5]. Furthermore, edge nodes typically lack comprehensive security measures, rendering them more susceptible to cyber-attacks. Thus, it is necessary to prevent potential privacy leakage in cloud-edge collaborative large model services.

In this paper, we propose a secure and efficient cloud-edge collaborative framework for federated large model services, by harnessing the computing power of edge nodes. Moreover, we propose three solutions to address the challenges in the corresponding framework. Specifically, we first devise a data-model-hybrid multi-level privacy protection mechanism to provide effective privacy guarantee in large model services at both the data and model levels. Afterward, we design a lightweight reciprocal knowledge transfer approach to achieve efficient large model training and deployment for resource-constrained and architecture-heterogeneous edge nodes. Lastly, we develop an optimal and reliable edge node recruitment strategy to efficiently recruit suitable edge nodes and exclude potential malicious nodes for targeted downstream task in a privacy-preserving way.

II. Architecture of Cloud-Edge Collaboration Framework

A. Workflow of Cloud-based Large Model Services

As depicted in Fig. 1, the workflow of the conventional large model service primarily comprises three stages: pre-training, fine-tuning, and deployment, predominantly conducted in cloud-based environments. The workflow commences with the pre-training of a foundation model (or called large model). Subsequently, the foundation model undergoes fine-tuning for specific downstream tasks, followed by using model compression and pruning techniques to generate multiple versions tailored to diverse service requirements. Finally, these trained models are deployed to offer intelligent services. The details of the three stages are as follows:

1) *Pre-training.* The foundation model pre-training stage utilizes a high-performance cloud computing cluster with significant computational capabilities. This stage primarily employs self-supervised learning algorithms to enable the pre-trained model to capture generic features. During this stage, a large-scale multi-modal dataset undergoes refinement in the cloud through data cleaning, augmentation, and fusion processes. Subsequently, the foundation model is trained on the refined dataset for enhanced capacity for feature representation and generalization performance.

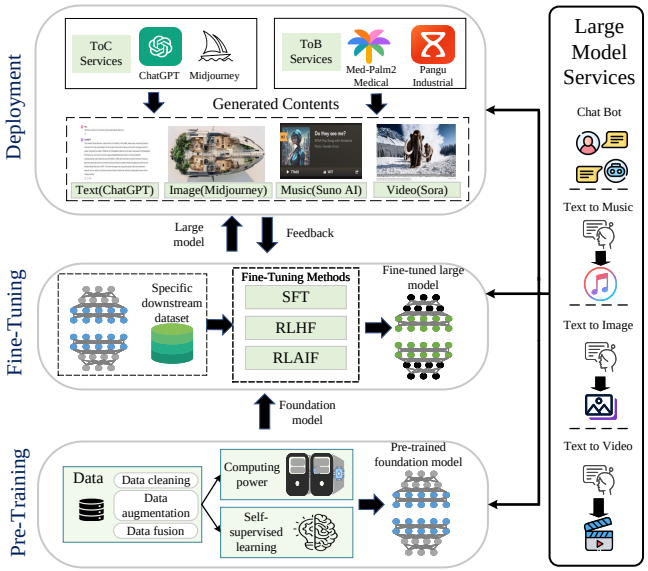

Fig. 1: Workflow of Conventional Cloud-based Large Model Services.

2) *Fine-Tuning.* To adapt the pre-trained foundation model for various downstream applications, it undergoes further fine-tuning using a targeted dataset, typically smaller and more specific compared to the pre-training dataset. The fine-tuning process involves three distinct methodologies: supervised fine-tuning (SFT), reinforcement learning from human feedback (RLHF), and reinforcement learning from AI feedback (RLAIF). SFT utilizes manually labeled data to refine the pre-trained model, whereas RLHF and RLAIF optimize the pre-trained model based on different types of feedback.
3) *Deployment.* Fine-tuned models such as ChatGPT and Midjourney are deployed into specific applications which may further undergo optimizations such as model compression and hardware adaptations. The services powered by large models can be classified into two types: 1) ToB (to business), where models are fine-tuned with institutional business data, and 2) ToC (to consumer), where models are fine-tuned with personal profiling data. Throughout the lifecycle of these services, user-generated data and feedback are continuously returned to the cloud servers, contributing to the refinement of both pre-training and fine-tuning stages.

B. Cloud-Edge Collaborative Framework for Large Model Service

Conventional cloud-based large model services suffer from high latency, significant communication overheads due to geographical distances and extensive IoT data. Cloud-edge collaboration can effectively alleviate these issues, but it

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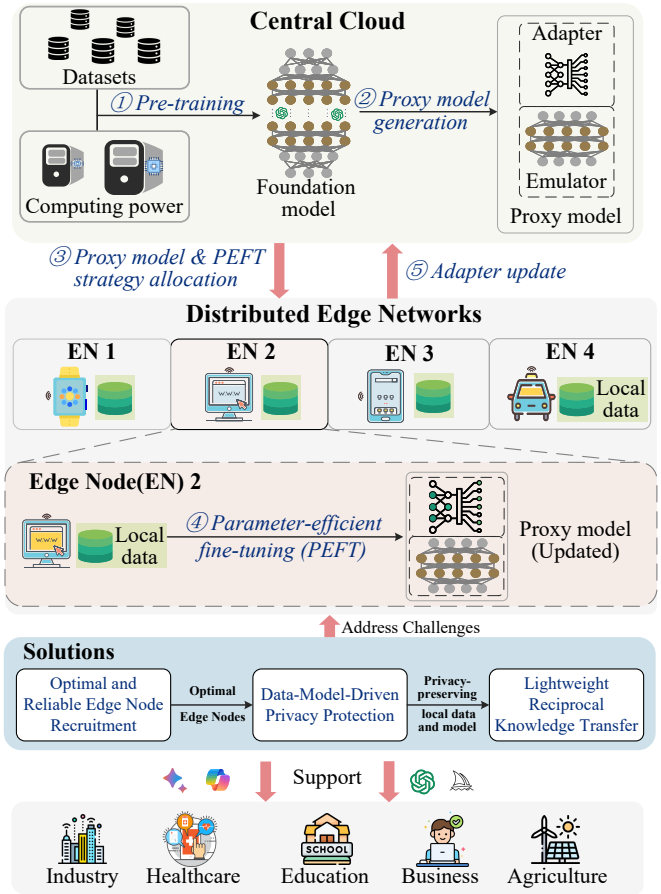

Fig. 2: Architecture of Cloud-Edge Collaboration for Large Model Services.

also faces challenges like model intellectual property, limited resources on edge nodes and privacy concerns.

As illustrated in Fig. 2, we present a cloud-edge collaborative framework to offer efficient and secure large model services, by leveraging the computation capability and local data resource of edge nodes. The construction of large model services in edge networks comprises two key processes: (i) training and (ii) deployment. Specifically, the federated large model training process encompasses the following three phases:

1) *Foundation model pre-training in cloud:* The foundation model pre-training process in the cloud aligns with conventional large model training process. Leveraging the powerful computational resources, the cloud aggregate sufficient public data to construct a foundation model capable of effectively capturing generic data features (step ①).
2) *Proxy model generation in cloud:* To avoid directly transmitting the original foundation model during the fine-tuning phase, the cloud employs the offsite-tuning transfer learning technique [6], which generates a proxy model comprising a lightweight adapter and a compressed emulator (step ②). The adapter serves as a plug-in for the foundation model, and encodes task-specific knowledge with a minimal number of parameters. Meanwhile, the emulator mimics the functions of the foundation model and supports the fine-tuning of the adapter by providing approximate gradient information.
3) *Federated model fine-tuning in edges:* edge nodes employ parameter-efficient fine-tuning (PEFT) techniques, such as Prefix-tuning [7] and LoRA [8], to further mitigate communication overheads and meet computational requirements. Initially, the cloud distributes the proxy model and the PEFT strategy to edge nodes (step ③). During each communication round of federated fine-tuning, the edge nodes train the PEFT module of the proxy model using their local data (step ④) and then transmit the updates to the cloud (step ⑤). Then, the cloud aggregates the updates and obtains the aggregation results to update the proxy model. Upon the completion of federated fine-tuning, the adapter is plugged into the foundation model to acquire the fine-tuned large model. Concurrently, the emulator undergoes fine-tuning by each edge node locally to facilitate collaborative service deployment.

The large model deployment process employs a cloud-edge coordinated model system. The aforementioned fine-tuned emulators are deployed in the edge nodes while the fine-tuned foundation model is deployed in the cloud. The response strategy is tailored to the users' requirements. Namely, for simple queries that demand quick responses, the emulator at the edge independently provides the answers; in contrast, for complicated requests where the response quality is paramount, the edge relays the requests to the cloud, then the fine-tuned model in the cloud analyzes the requests and delivers comprehensive responses. To further implement a secure and efficient cloud-edge collaborative framework for large model services, we propose solutions to address key challenges in Sec. III, including data memorization privacy threat in large models, architecture-heterogeneous and resource-constrained edge nodes, and unreliable edge nodes with heterogeneous data distribution. The workflow of the proposed solutions are illustrated in the bottom part of Fig. 2. Specifically, we first employ an optimal and reliable edge node recruitment strategy to select the most suitable edge nodes for specific downstream tasks. Subsequently, a data-model-driven privacy protection mechanism is devised to ensure the privacy of the edge nodes' local sensitive data. Finally, efficient large model federated fine-tuning is achieved through a lightweight reciprocal knowledge transfer approach tailored for resource-constrained and architecture-heterogeneous edge nodes.

In summary, the proposed cloud-edge collaborative large model service offering framework enjoys the following benefits:

- *Efficiency*: The communication and computational overheads can be significantly reduced during federated fine-tuning process by leveraging proxy model and PEFT techniques within the cloud-edge collaborative framework.
- *Scalability:* By using a unique adapter plug-in mechanism to extend the functionality of the foundation model, the cloud-edge collaborative framework facilitates the expansion of the foundation model to diverse downstream

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tasks.

- *Privacy Protection:* The cloud-edge collaborative framework ensures the privacy of both cloud and edge nodes by eliminating the need of sharing sensitive local data or the cloud's original foundation model.

III. Challenges of Secure and Efficient Cloud-Edge Collaborative Framework

A. Data Memorization and Privacy Leakage of Large Models

The data memorization phenomenon in large models is an emerging privacy concern, as the large models generally have extensive learning capabilities to retain detailed information from training datasets. Recent research has demonstrated that large language models (LLMs) can retain sensitive training data, posing a risk for user privacy leakage through model outputs [4], [5]. For instance, it is reported that ChatGPT users inadvertently accessed payment details from other users' interactions, including names, email addresses, and credit card numbers. However, existing privacy-preserving schemes primarily focus on preventing privacy attacks from the model perspective and can not effectively prevent potential data leakage stemming from data memorization [9], [10]. In the context of cloud-edge collaborative large model training, edge nodes utilize local sensitive data (such as location and health information) to fine-tune the proxy model, which can inadvertently lead to data memorization issues and the corresponding privacy breaches. Hence, it is imperative to develop privacy enhancement techniques to enhance the privacy of large models from both model and data perspectives.

B. Resource-Constrained Edge Nodes with Diverse Processor Architectures

The cloud-edge collaborative framework effectively alleviates both communication and computation overheads during the federated fine-tuning process [11], [12]. However, even with the use of compression techniques including proxy models and PEFT, part of resource-constrained edge nodes still cannot afford the computational cost and storage requirements for training and deploying large models.

Furthermore, resource-constrained edge nodes are often operated in diverse processor architectures (e.g., x86, ARM architectures), making it challenging to uniformly fine-tune or deploy models across them. In these cases, the resource-constrained edge nodes have to transmit their local data to more capable edge nodes, which may raise privacy concerns and reduce the intentions of edge nodes to participate in federated fine-tuning. Therefore, it is essential to develop a lightweight and architecture-free fine-tuning approach to involve more edge nodes with limited resources and various processor architectures.

C. Unreliable Edge Nodes and Data Heterogeneity

Compared with the cloud, the heterogeneous edge nodes can be more easily to be compromised by adversaries and exploited to perform various attacks (e.g., poisoning attacks, and backdoor attacks) to deteriorate the quality of large models or to inject backdoors. Consequently, it is crucial to verify the reliability of edge nodes and ensure that they contribute benign updates during the federated fine-tuning process. Previous research on node recruitment and malicious node exclusion in FL typically requires nodes to submit their actual local model updates in each communication round [13], which raises efficiency issue in the scenarios of large model services. Moreover, raw updates uploads may raise privacy concerns such as gradient inversion attacks [14]. However, due to the high heterogeneity of data across diverse edge nodes, it poses a significant challenge for the cloud to accurately evaluate the contribution of edge nodes in non-IID scenarios. Thereby, it remains an open issue to select reliable devices for specific downstream task to resist compromised or malicious edge nodes in federated large model fine-tuning.

IV. Solutions to Secure and Efficient Cloud-Edge Collaborative Framework

A. Data-Model-Driven Privacy Protection Mechanism

In this subsection, as illustrated in Fig. 3, we present a data- and model-driven privacy protection mechanism to efficiently prevent private data memorization threats during federated large model training. This mechanism enhances the privacy preservation capability of the proposed framework at both the data and model levels, which can effectively defend against privacy inference and extraction attacks tailored to large model services. By avoiding the utilization of complex cryptography techniques, the proposed framework circumvents the computational limitations of edge nodes. Furthermore, it offers personalized model-level privacy protection tailored to various privacy requirements, enhancing its flexibility and applicability in real-world scenarios. Specifically, 1) At the data level, privacy is ensured through two key modules: *data desensitization* (by masking sensitive information in the original local dataset) and *data augmentation* (by generating alternative and non-sensitive data samples). 2) At the model level, the *local differential privacy (LDP)* and *sign-only gradient uploading* mechanisms are utilized to meet diverse privacy requirements. Notably, data-level and model-level privacy protection methods can be utilized either jointly or separately.

1) Data-level privacy protection. It consists of the following three stages. *(i) Sensitive information detection.* Each edge node runs the designed binary sensitive information detection model to identify potentially sensitive data samples (e.g., names or phone numbers). The data samples in the local dataset are classified as sensitive and non-sensitive ones. Subsequently, a multi-label classification model is designed to further categorize the sensitive data samples more precisely. For instance, the sensitive data samples can be categorized into data samples with location/identification/health information, and so on. *(ii) Mask-based data desensitization.* Edge nodes employ symbol masking to replace sensitive data elements in the private data. For instance, phone numbers may be transformed into a series of asterisks. *(iii) Alternative data generation.* Conditional generative adversarial networks (cGANs) are utilized to generate alternative data similar to the real local

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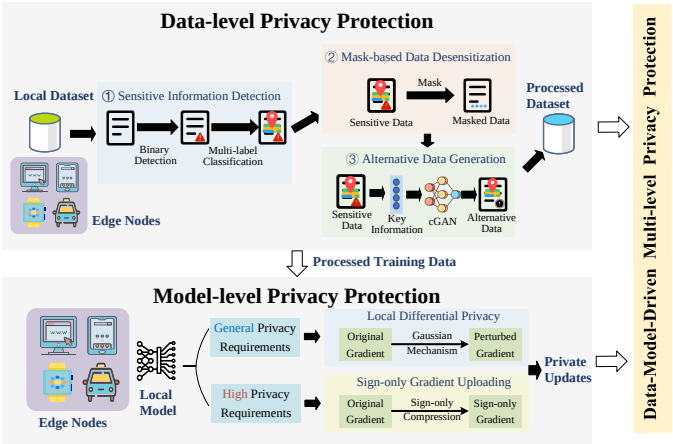

Fig. 3: An illustration of Data-Model-Driven Privacy Protection Mechanism.

dataset. We further devise a information bottleneck-based (IB-based) data generation mechanism in conventional cGANs, utilizing the key information of original data to enhance the quality of synthetic alternative data. The key information is generally the most relevant features of the target downstream task without sensitive information. Specifically, the key information can be utilized as additional conditioning information in cGANs, which ensures the generator and discriminator focus on these information when generating alternative data samples.

2) Model-level privacy protection. It can be categorized into two types based on the specific privacy requirements of the edge nodes. (i) For edge nodes with general privacy requirements, the LDP mechanism is employed. Specifically, edge nodes utilize the Gaussian mechanism to generate noise aligned with the privacy budget, which is then added to the original gradient for perturbation during the federated model fine-tuning phase. (ii) For edge nodes with higher privacy requirements, the sign-only gradient uploading approach is utilized by only transmitting the sign of the gradient to the cloud when fine-tuning the foundation models. As such, it enhances privacy protection within edge networks.

B. Lightweight Reciprocal Knowledge Transfer Approach

In this subsection, as illustrated in Fig. 4, we devise a lightweight reciprocal knowledge transfer approach for efficient large model services among resource-constrained edge nodes. The approach utilizes the knowledge of edge nodes, such as soft predictions or high-level features, to guide fine-tuning of foundation models for downstream tasks. Data collected during large model deployment is transferred from cloud to edge for knowledge distillation on local models, improving their performance and enabling reciprocal knowledge transfer. Unlike conventional methods, only a minimal amount of data and knowledge are transferred, reducing communication overheads and computational requirements significantly. The workflow of the proposed approach mainly includes four phases. Firstly, the cloud generates a fine-tuning dataset by collecting or synthesizing data. Secondly, it allocates the dataset to edge nodes and extracts their local model knowledge. Thirdly, the cloud fuses edge node knowledge to annotate the fine-tuning dataset and fine-tune the foundational model by reverse knowledge transfer. Finally, the cloud collects user input-output pairs to create a distillation dataset after large model deployment, periodically enhancing edge node models by knowledge distillation. In this way, edge nodes with limited resources and heterogeneous architectures can be involved in the proposed cloud-edge collaborative large model framework, thereby introducing more downstream data and improving the performance of large models.

- **Phase 1: Fine-tuning Dataset Generation.** In this phase, the cloud generates a fine-tuning dataset whose data distribution aligns with the target downstream task, to extract knowledge from the edge nodes. Firstly, the cloud collects publicly available data from the Internet. Secondly, it employs both the public data and a diffusion-based model to synthesize the additional data. The fine-tuning dataset comprises a combination of both public and synthetic data.
- **Phase 2: Knowledge Extraction.** In this phase, each edge node trains an AI model with resource-friendly architectures (e.g., MobileNet and ShuffleNet) using local data. The cloud then allocates the fine-tuning dataset to a group of selected edge nodes. These edge nodes extract knowledge by feeding the fine-tuning dataset into their local models, and subsequently upload the knowledge back to the cloud.
- **Phase 3: Reverse Knowledge Transfer.** In this phase, the cloud deploys a knowledge fusion module to fuse the knowledge from the edge nodes. Specifically, the plurality voting or soft-prediction weighted aggregation methods are utilized to generate the unified knowledge set, and then the fused knowledge is leveraged to annotate the fine-tuning dataset. The foundation model is trained on the fine-tuning dataset for several epochs to achieve the reverse knowledge transfer from edge node models to the foundation model. Notably, in this scenario, the foundation model is considered the "student", while the edge node models act as "teachers", reversing the conventional knowledge transfer roles.
- **Phase 4: Periodical Knowledge Distillation in Edge Nodes.** This phase is optional in our proposed approach, depending on the privacy requirements of downstream tasks. Case 1: For downstream tasks with lower privacy requirements, service providers can periodically collect user inputs and the corresponding outputs from the large model service and forms a small-amount distillation dataset, provided that user consent is obtained and relevant regulations are followed. Subsequently, the distillation dataset undergoes data desensitization to eliminate any private information before being allocated to edge nodes. The edge nodes then improve their resource-friendly local models via knowledge distillation utilizing the allocated dataset, thereby they can better assist the subsequently periodical large model fine-tuning with perceived new local data. Case 2: For more sensitive

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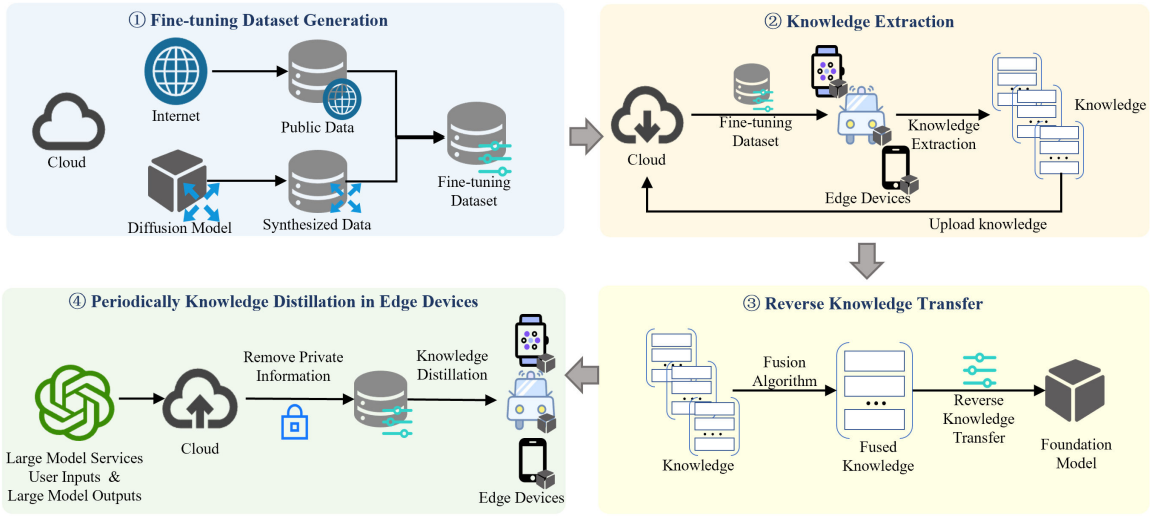

Fig. 4: An illustration of Lightweight Reciprocal Knowledge Transfer Approach.

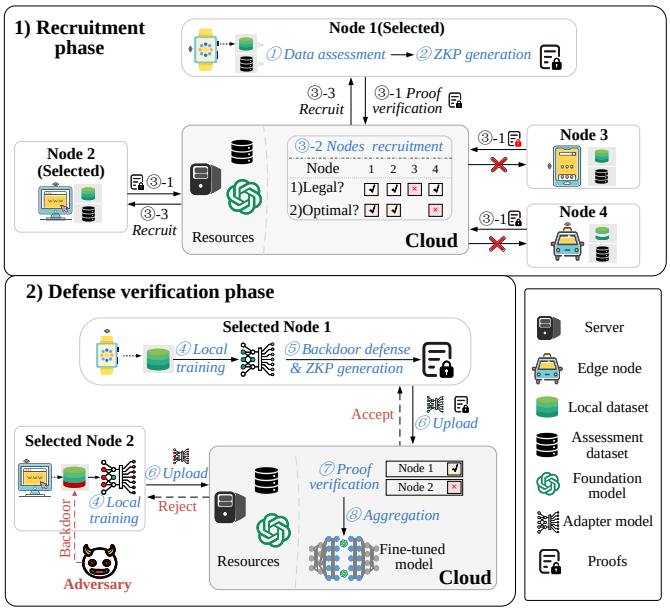

Fig. 5: An illustration of Optimal and Reliable Edge Node Recruitment Strategy under Non-IID Environment.

downstream tasks with a higher frequency of privacy concerns, service providers should avoid collecting user input-output pairs, and cloud-to-edge knowledge transfer should not be performed. The Kullback-Leibler (KL) divergence serves as the loss function, quantifying the discrepancy between the probability distributions predicted by local models and those predicted by the fine-tuned large model.

C. Optimal and Reliable Edge Node Recruitment Strategy

In this subsection, we propose an innovative strategy for recruiting optimal and reliable edge nodes, by leveraging the data assessment to handle data heterogeneity issues and the defense verification to mitigate the risk of potential malicious nodes. The proposed recruitment strategy enables efficient quality assessment with the aid of a small prepared assess dataset, reducing the communication overheads of edge nodes in cloud-edge collaborative large model training process. Additionally, the proposed strategy eliminates privacy concerns by employing zero-knowledge proofs (ZKPs) to verify the integrity of edge nodes without uploads of local data or raw updates. ZKP is a cryptographic technique enabling one party, known as the prover, to convince another party, known as the verifier, that a statement is true without disclosing any information beyond the validity of the statement. In this paper, edge nodes integrate the correctness of assessment (or defense method execution for defense verification) with a random value to establish the secret function of ZKPs. The Pederson commitment is utilized as the commitment scheme. As shown in Fig. 5, the recruitment strategy consists of two phases: (i) recruitment phase and (ii) verification phase.

1) Recruitment Phase: Edge nodes assess their data quality for specific tasks, and then generate ZKPs of these assessment results. The cloud then verifies these proofs, computes data quality scores, and recruits the optimal edge nodes. Specifically, this phase can be divided into the following three steps:

- *Data assessment.* Edge nodes receive a specially prepared assessment dataset from the cloud, which involves a small number of collected data samples from the downstream environment. Then edge nodes compute the Earth mover's distance (EMD) between the label distribution of the assessment dataset and their local data as the assessment metric of data quality.
- *Proof generation.* Then, edge nodes generate ZKPs regarding the correctness of the assessment. These proofs, along with the EMD and dataset size, are then transmitted to the cloud.
- *Proof verification and recruitment.* The cloud verifies the proofs, excluding the devices with illegal proofs. Subsequently, it computes the data quality scores of devices based on the EMD and dataset size, and recruit

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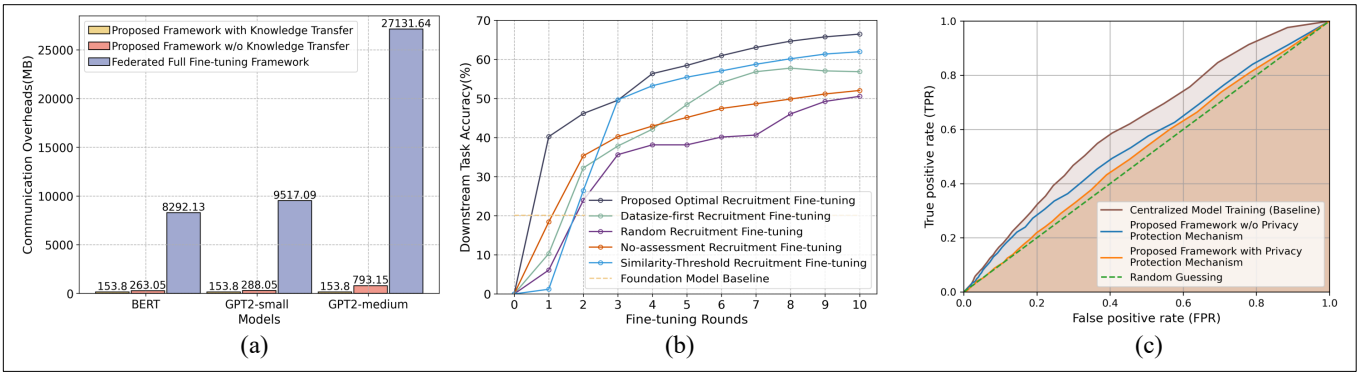

Fig. 6: (a) Communication overheads during the federated fine-tuning process. (b) Fine-tuning performance of different recruitment strategies. (c) Defense capabilities against membership inference attacks.

the top-K edge nodes with the highest data quality scores.

2) Defense Verification Phase: In this phase, the cloud and edge nodes follow the secure aggregation protocol to guarantee the privacy protection. After local training, edge nodes implement defense mechanisms against backdoor attacks on the original updates, such as pruning defense, and generate ZKPs to demonstrate the execution of these defenses. The cloud then receives the perturbed updates and proofs from edge nodes, thereby verifying the proofs to identify any potential malicious device. Notably, potential malicious devices are regarded as "dropped users", and their perturbations are reconstructed by leveraging the secret shares from benign devices. Finally, the cloud aggregates the benign updates while discarding perturbations from potential malicious devices to derive the aggregated update. The aggregated update is then utilized to refine the adapter model and fine-tune the foundation model.

V. Case Study

The case study is carried out on a workstation with an Intel Xeon Platinum 8280 CPU (2.7GHz/4.0GHz), 256G RAM, and two Nvidia GeForce RTX 3090 GPUs. PyTorch is utilized to implement the cloud-edge collaborative large model training framework, involving 20 edge nodes as participants. The foundation models considered include BERT, GPT2-small, and GPT2-medium. The target downstream task is a text classification task using the 5-classes Yelp Review dataset[2]. Each edge node has varying amounts of local data; 10 nodes possess 200 samples, while the rest have 100. Additionally, 10 nodes only have samples from classes $[0, 1]$, while the others have classes $[2, 3, 4]$. We conduct 10 fine-tuning rounds, employing LoRA [8] as the PEFT technique. The local learning rate is set to 5^{-5}, and the local batch size is set to 8. We have a test dataset of 6000 samples from classes $[0, 1]$, which is also used for downstream task evaluation. The membership inference attacks are implemented to illustrate the privacy protection performance of the proposed framework.

Fig. 6(a) illustrates communication overheads among the proposed framework with knowledge transfer, the proposed framework without knowledge transfer, and the original full fine-tuning framework [15]. For the *proposed framework with knowledge transfer*, only the knowledge of edge nodes and cloud are transmitted, which is consistent across all model architectures. For the *framework without knowledge transfer*, the proxy model and the gradients of PEFT modules are transmitted. For the original *full fine-tuning framework*, the original foundation model and corresponding gradients are transmitted. As observed in Fig. 6(a), the proposed framework significantly reduces the communication overheads while maintaining privacy, making it effectively applicable for resource-constrained edge nodes.

Fig. 6(b) illustrates the fine-tuning performance of 5 edge node recruitment strategies using GPT-2 small as the foundation model, with data distribution among edge nodes being highly non-IID. In the experiments, we compare the proposed optimal recruitment strategy with other four conventional recruitment strategies. As seen in Fig. 6(b), the proposed optimal recruitment strategy outperforms other strategies in terms of the accuracy of downstream task. The reasons are as follows. The *datasize-first recruitment strategy* selects the top-K edge nodes solely based on local data size. The *similarity-threshold recruitment strategy* recruits edge nodes solely based on the similarity (measured by Euclidean distances) between data distributions, selecting nodes only when the similarity is below a specific threshold. The *random recruitment strategy* selects K edge nodes randomly. The *no-assessment recruitment strategy* recruits all edge nodes without data assessment. The first two strategies focus solely on data size or distribution, potentially overlooking outstanding edge nodes that balance both factors. In contrast, the latter two strategies may select edge nodes with data distributions that are inconsistent with downstream needs, resulting in performance degradation. The proposed optimal recruitment strategy recruits the top-K edge nodes according to the quality scores based on the EMD and local data size, thereby enjoying the highest task accuracy.

Fig. 6(c) illustrates the defense performance of the proposed framework against membership inference attacks. In the experiments, we utilize the *area under curve (AUC)* of the receiver operating characteristic (ROC) curve as the privacy-preserving

[2]https://huggingface.co/datasets/yelp_review_full

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metric under membership inference attacks. The smaller the AUC, the lower the success rate of membership inference attacks. As seen in Fig. 6(c), it demonstrates that the proposed framework effectively reduces the success rate of such attacks against large models. Additionally, the proposed hybrid data-model-driven privacy protection mechanism further enhances the defense capabilities, bringing the success rate of membership inference attacks on cloud-edge collaborative large model services close to the level of random guessing.

VI. Future Research Directions

A. Federated Large Model Ownership Verification

In edge networks, fine-tuning a capable large model necessitates substantial computational and communication resources. Federated large models, requiring collaborative fine-tuning across edge nodes, pose ownership issues. Model watermarking offers a solution by embedding private watermarks into AI models, safeguarding the intellectual property (IP) rights. However, current model watermarking schemes are not suitable for federated large models in edge networks. Therefore, novel mechanisms are needed for ownership verification and IP authentication in this context.

B. Semantic Communication-enabled Edge Intelligence

Semantic communication technology reduces communication overheads for large model services by transmitting only essential, semantically relevant information between edge nodes and servers. Integrating it with the cloud-edge collaborative framework enhances the fine-tuned large model performance and saves communication bandwidth. Specifically, semantic communication can be adapted to refine communication protocols within edge networks for specific applications like urban traffic management and personal health monitoring. Further research efforts are required to design an efficient semantic communication-enabled large model training framework.

C. Personalized and Customized Large Model in Edge Networks

The rise of deploying large personalized models on edge devices marks a growing trend. However, the complexity lies in diverse user requirements and data heterogeneity, posing challenges for personalized services like virtual personal assistants. To tackle this, it is necessary to conduct fine-grained fine-tuning to obtain various versions of fine-tuned model for diverse users. Furthermore, the user-specific profiles can be utilized for personalization fine-tuning by pre-inputting prompts.

VII. Conclusion

In this paper, we propose a novel cloud-edge collaborative framework for large model services across edge networks. The proposed framework can harness the edge resources and significantly reduce communication overheads. We propose three innovative solutions to address the key challenges within this framework: a data-model-driven privacy preservation mechanism to prevent private data memorization, a lightweight reciprocal knowledge transfer approach to efficiently deploy large models on resource-constrained edges, and an optimal and reliable edge node recruitment strategy under data heterogeneity to exclude malicious edge nodes. Furthermore, we provide a case study to validate the effectiveness of the proposed framework and solutions. Finally, open and essential future research directions are discussed.

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